

Experiment 6: 5G NR Frame Structure Analysis with Different Subcarrier Spacing

Objectives, MATLAB Programs and Output Images

Objective 1

To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology.

MATLAB Program

```
clc;
clear;
close all;

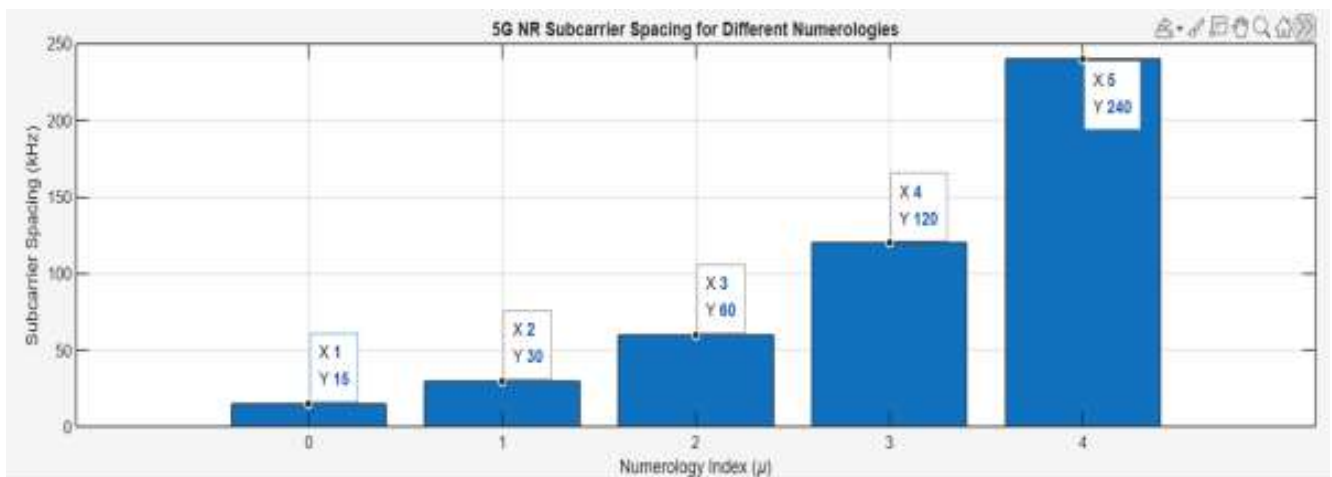
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
mu = 0:4; % Numerology Index

bar(scs)
grid on

set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)

xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```

Output Image – Objective 1



Objective 2

To compare the slot duration corresponding to different Subcarrier Spacing values in 5G NR.

MATLAB Program

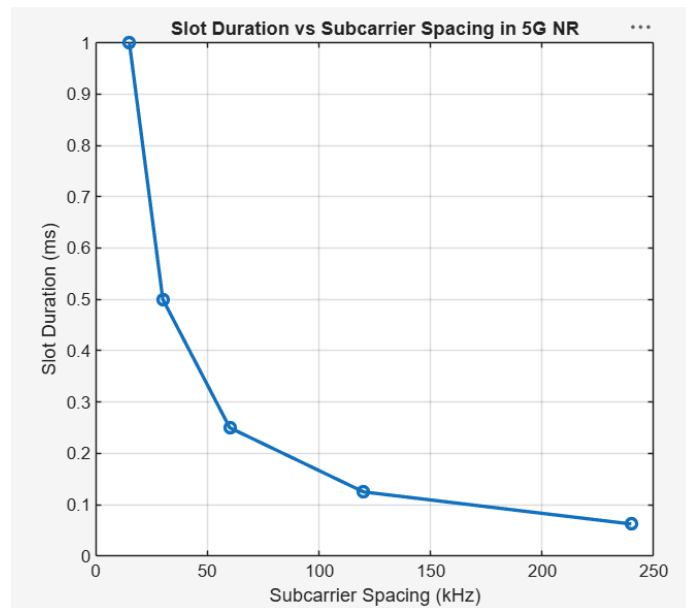
```
clear;
close all;

scs = [15 30 60 120 240];           % Subcarrier Spacing (kHz)
slot = [1 0.5 0.25 0.125 0.0625];   % Slot Duration (ms)

plot(scs, slot, 'o-', 'LineWidth', 2)
grid on

xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')
```

Output Image – Objective 2



Objective 3

To study the number of OFDM symbols present in a 5G NR slot for normal cyclic prefix.

MATLAB Program

```
clc;
clear;
close all;

scs = [15 30 60 120 240];          % Subcarrier Spacing (kHz)
symbols = [14 14 14 14 14];        % OFDM Symbols per Slot

figure
bar(scs, symbols)

grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('OFDM Symbols per Slot for Different Subcarrier Spacing')

ylim([0 16])
```

Output Image – Objective 3

